

Coding & Solution Template

Date	15 March 2026
Team ID	Batch 2023-27 ACOE
Project Name	Rising Waters – ML-Based Flood Prediction System
Maximum Marks	3 Marks

7-Step ML Implementation Pipeline

Step	Activity	Key Actions	Output
1	Environment Setup	Install Python, Flask, Scikit-learn, XGBoost, Pandas	requirements.txt
2	Dataset Collection	Generate 5,000-row synthetic dataset with 6 features	flood_dataset.csv
3	Data Visualisation	EDA: distribution plots, box plots, pairplot, heatmap	4 EDA charts
4	Data Preprocessing	Mean imputation, IQR outlier capping, StandardScaler	scaler.save
5	Model Building	Train Decision Tree, Random Forest, KNN, XGBoost	4 trained models
6	Best Model Selection	Compare accuracy: XGBoost wins at 96.55%; serialize	floods.save
7	Web App + Deployment	Flask app with 6 pages; Docker containerize; deploy	Live web app

Flask Application Routes

Method	Route	Description	Template
GET	/	Home Dashboard	home.html
GET	/predict	Input Form	predict.html
POST	/predict	Process form → validate → scale → predict → redirect	—
GET	/result/flood	Flood Chance — emergency action cards	result_flood.html
GET	/result/noflood	No Flood — monitoring recommendations	result_noflood.html
GET	/history	Prediction audit log from predictions_log.csv	history.html
GET	/about	Architecture, model info, team details	about.html